SAMARTH

CUET PG 2027

Lecture – 13 | Computer Science

Unit 03 : Digital Fundamental

Topic : Sequential Circuits

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Topic : Sequential Circuits

Examples.

1. Latches.
2. Flip Flops
3. Registers
4. Counters.

Recap *of previous lecture*

1. Combinational Circuits

- ALU
  - Adder
  - Subtractor
- Data Comm.
  - MUX
  - DeMUX
  - Encoder
  - Decoder
- Code converters



Topics *to be covered*

1. Sequential Circuits

Q. Which of the following are SC?

① Flip Flop — SC
② MUX — CC
③ Counter — SC
④ Decoder — CC
⑤ Adder — CC

(1) 2, 4
(2) 1, 3
(3) 3, 5
(4) 1, 4



Topic : Sequential Circuits

Ex AC Timer.

Ex Pin "1234"

→ CC + Memory.

Sequential circuits are digital circuits that store and use previous state information to determine their next state. They are commonly used in digital systems to implement state machines, timers, counters, and memory elements and are essential components in digital systems design.

Vending Machine

A combinational circuit produces an output based on input variables only, but a **sequential circuit** produces an output based on **current input and previous output variables**. That means sequential circuits include memory elements that are capable of storing binary information. That binary information defines the state of the sequential circuit at that time. A latch capable of storing one bit of information.

Ex Gates.

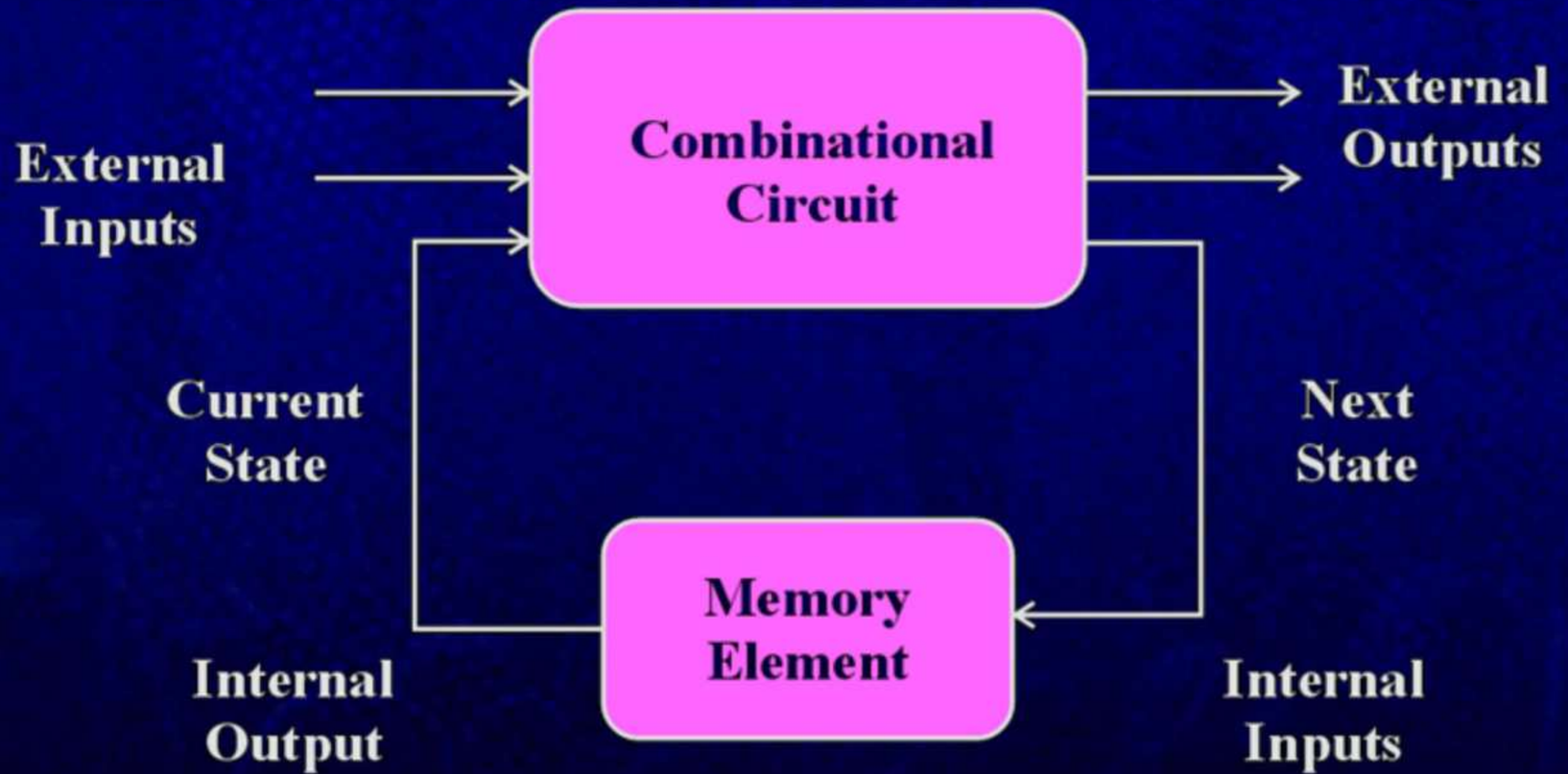
Topic : Sequential Circuits



Figure: Combinational Circuits

Topic : Sequential Circuits

Sequential circuit



Topic : Sequential Circuits

There are **two types** of input to the **combinational logic**:

1. **External inputs** which are not controlled by the circuit.
2. **Internal inputs**, which are a function of a previous output state.

Secondary inputs are state variables produced by the storage elements, whereas secondary outputs are excitations for the storage elements.

Topic : Sequential Circuits

There are **two types of Sequential circuits**:

1. **Synchronous sequential circuits:** These circuits **uses clock signal** and level inputs (or pulsed) (with restrictions on pulse width and circuit propagation). The output pulse is the same duration as the clock pulse for the clocked sequential circuits. Since they wait for the next clock pulse to arrive to perform the next operation, so these circuits are bit **slower** compared to asynchronous. Level output changes state at the start of an input pulse and remains in that until the next input or clock pulse.

Ex CPU.
(Instruction cycle
, decode, Execute).

Ex Musician.

Ex Train Time.

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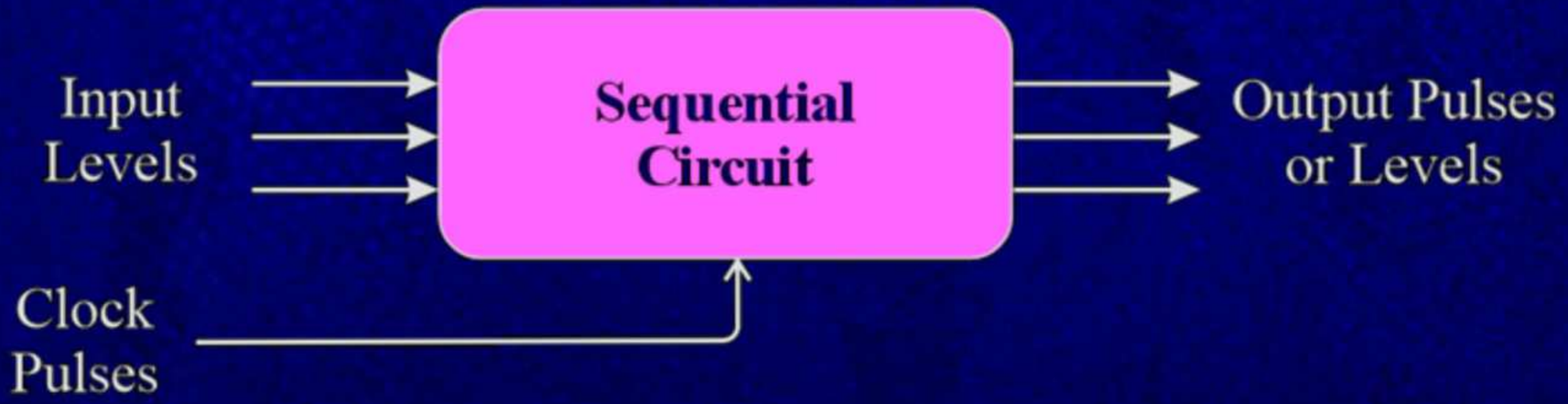


Figure: Synchronous Sequential Circuit

2. **Asynchronous sequential circuits:** These circuits do not use a clock signal but uses the pulses of the inputs. These circuits are faster than synchronous sequential circuits because there is no clock pulse and change their state immediately when there is a change in the input signal.

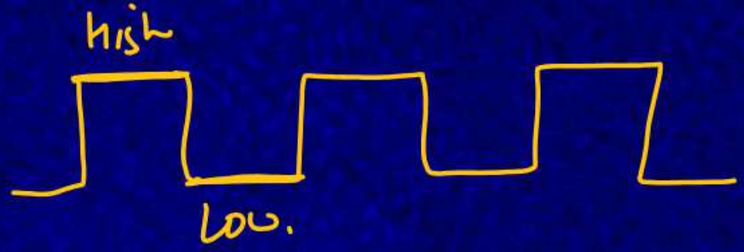
Ex PC Keyboard

Ex Private Taxi



Figure: Asynchronous Sequential Circuit

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Clock Signal and Triggering

Clock signal is a kind of control signal that allows the elements of synchronous circuits to be in phase or phenomena that occur in circuits. It is derived from the square wave that has a high and a low level, it helps in measuring the sequential changes in the circuit states. The clock signal also makes a pulse simultaneously on all the circuit parts that are needed for the proper work of synchronous sequential circuits.

In Sequential circuits, triggering denotes the way, in terms of which the state changes take place. There are two main types of triggering:

↓ give commands to memory element (Latches, Flip Flops)

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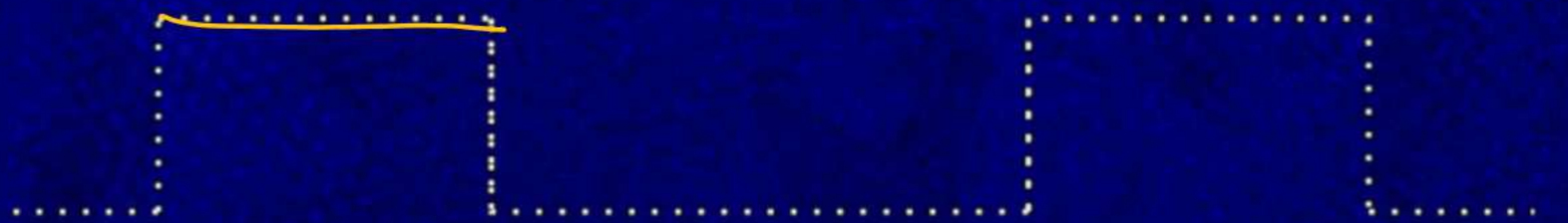


Level Triggering

Level triggering happens when the change of state is from the level of the clock signal is high or low. The circuit depends on the level of the clock signal rather than the rising or the falling edge of it. There are two types of level triggering:

1. **Positive Level Triggering:** The circuit changes state when it is high time in the clock cycle i.e. when the clock signal is high.

2. **Negative Level Triggering:** The circuit changes state when the clock signal is in the low state.



Positive Level Trigger



Negative Level Trigger

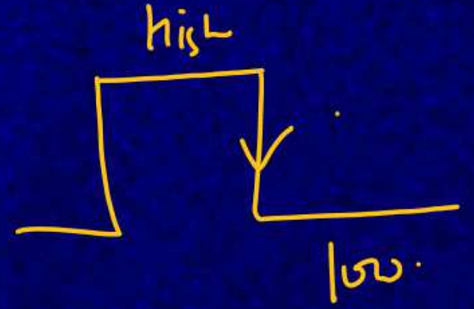
Edge Triggering

Edge triggering occurs when the state change is initiated by the transition (rising or falling edge) of the clock signal. The circuit responds to the clock signal's edges rather than its levels. There are two types of edge triggering:

1. **Positive Edge Triggering:** The circuit changes state on the rising edge (transition from low to high) of the clock signal.



2. **Negative Edge Triggering:** The circuit changes state on the falling edge (transition from high to low) of the clock signal.





Positive Edge Trigger



Negative Edge Trigger

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Q. Which of the following is true?

(1) O/P of CC depend only on current input. ✓

(2) O/P of SC depend on current input. ✗

(3) O/P of CC depend on current input & previous O/P. ✗

(4) O/P of SC " " both I/I & " " ✓

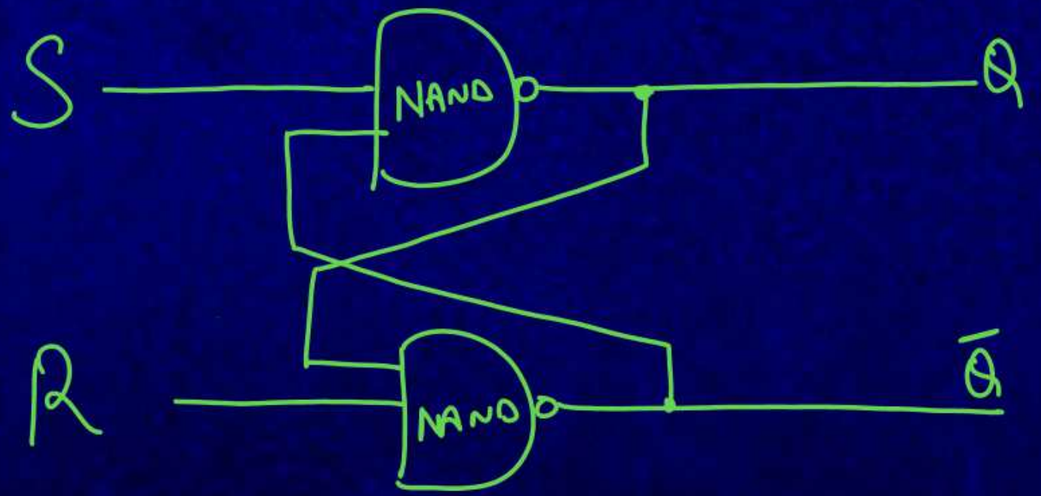
(1) 1, 3 (3) 3, 4

(2) 1, 2 (4) 1, 4

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Latches :- Basic building block.

SR latch using NAND gate.



NAND gate :

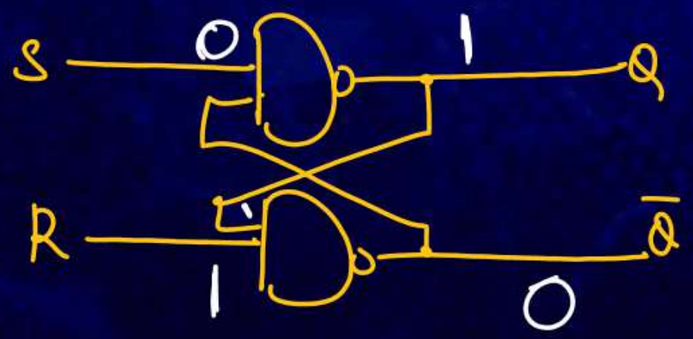
A	B	A.B	$\overline{A.B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

If any one input is 0, then O/P will be 1.

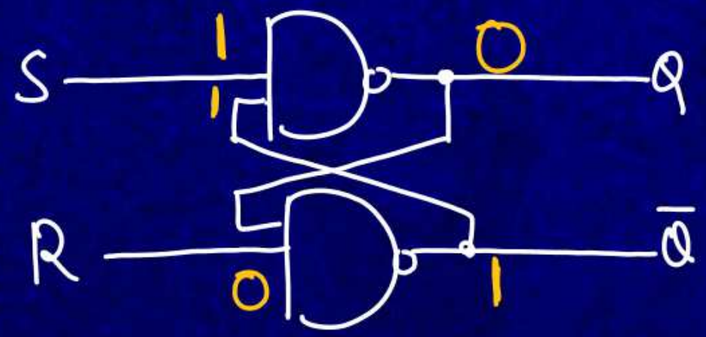
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S	R	Q_n
0	0	Invalid state.
0	1	1
1	0	0
1	1	Hold state.

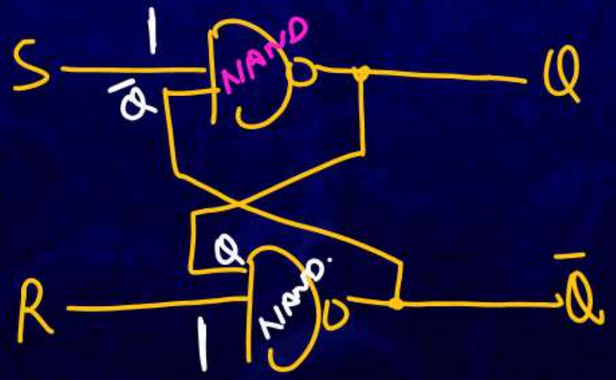
① $S=0, R=1$



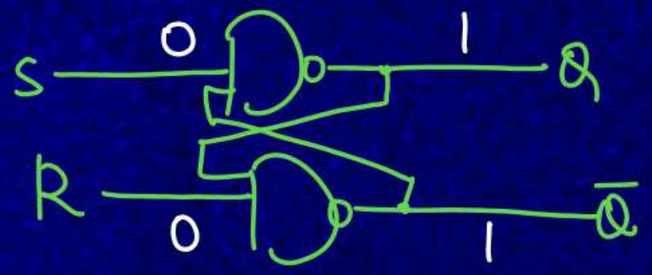
② $S=1, R=0$.



③ $S=1, R=1$



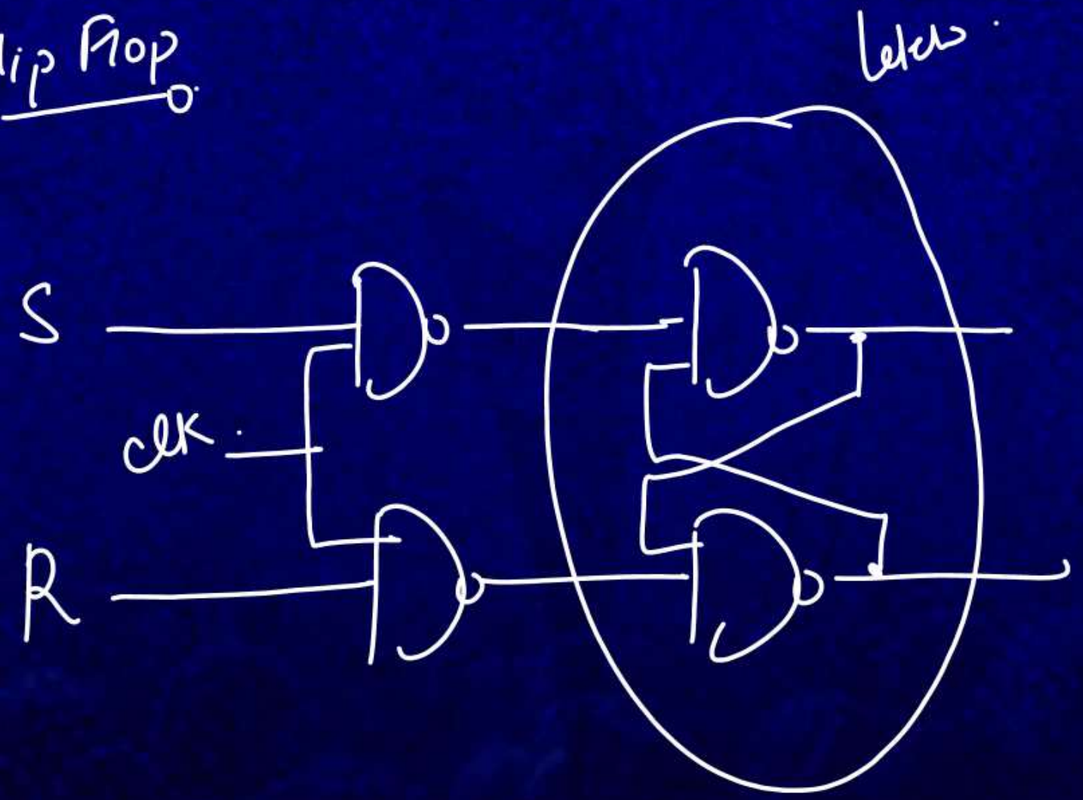
④ $S=0, R=0$.



$\overline{1 \cdot \bar{Q}} = \bar{1} + \bar{\bar{Q}} = 0 + Q$
$= Q$

$\overline{1 \cdot Q} = \bar{1} + \bar{Q} = 0 + \bar{Q}$
$= \bar{Q}$

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2 Mins Summary

- Sequential circuit
  - Basics
    - SR Latches

Homework from SAMARTH CUET PG 2027 Module

Revision + DPP.

THANK YOU